ECE186 Final Project
Detecting Market Regimes with the Baum-Welch and Viterbi Algorithms

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Abstract

We classify each trading day of the S&P 500 between 2010 and 2026 into a bull, bear, or sideways market regime using only the price history of SPY. Daily log returns are modeled as a Gaussian-emission Hidden Markov Model (HMM). The transition matrix, initial distribution, and emission parameters are estimated with the Baum-Welch algorithm, and the most likely regime sequence is then recovered with the Viterbi algorithm. The resulting classifications identify the COVID-19 crash of 2020 and the 2022 inflation-driven downturn as bearish regimes, and a comparison of two- and three-state models shows that the sideways state acts as a transition buffer between bullish and bearish periods.

1 Introduction

1.1 Objective

The goal of this project is to classify the regime of the stock market on each day between 2010 and 2026 as bear, bull, or sideways. Market returns are estimated using SPY, the State Street SPDR S&P 500 ETF Trust, a widely used S&P 500 tracker. As is common practice in finance, these returns can be modeled as a Hidden Markov Model.

1.2 Modeling Assumptions

This model is not a fully accurate description of the market, since returns and the current regime depend on more than the returns and regime of the previous day. We adopt the first-order Markov assumption so that the problem is tractable. It is also reasonable to assume that the best single indicator of the current regime is the preceding regime.

The Baum-Welch algorithm is used to estimate the transition parameters (the transition matrix A and the initial distribution π) together with the emission distribution parameters ($\mu_{\text{bull}}, \sigma_{\text{bull}}, \mu_{\text{bear}}, \sigma_{\text{bear}}$, and the corresponding sideways parameters in the three-state model). The transition parameters describe the relationships between regimes throughout the time series, while the emission parameters determine the probability of observing a given day's return under each regime. Once the parameters are estimated, the Viterbi algorithm recovers the most likely sequence of regimes over the full period.

1.3 Data

The price history of SPY is obtained from a public dataset. From the daily prices p_t we compute log returns,

$$r_t = \log\left(\frac{p_t}{p_{t-1}}\right), \tag{1}$$

where p_t is the price on day t and p_{t-1} is the price on the prior day. Log returns are standard in finance because they are additive over time: the return from day t to day $t+N$, for a positive integer N, is the sum of the N intervening daily log returns. The resulting series contains $T = 4131$ observations.

2 Methodology

2.1 Model Specification

Let $X_t \in \{1, \ldots, N\}$ denote the hidden regime on day t and $Y_t = r_t$ the observed return. The HMM is parameterized by $\theta = (A, \pi, \mu, \sigma)$, where A is the transition matrix with entries $a_{ij} = P(X_{t+1} = j \mid X_t = i)$, π is the initial state distribution, and each regime i emits returns from a normal distribution,

$$b_i(y) = \mathcal{N}(y; \mu_i, \sigma_i^2). \tag{2}$$

The transition matrix gives the probability of moving from one regime to another, and the initial distribution gives the starting regime. The emission parameters map the returns observed in each regime to a normal distribution with its own mean and variance. Both components are required because we must learn how regimes evolve from one day to the next as well as what returns are typical within each regime. Parameters are initialized with an educated guess based on the observed returns.

2.2 Forward-Backward Algorithm

Parameter estimation begins with the forward-backward algorithm, which computes

$$\alpha_i(t) = P(Y_1 = y_1, \ldots, Y_t = y_t, X_t = i \mid \theta), \tag{3}$$
$$\beta_i(t) = P(Y_{t+1} = y_{t+1}, \ldots, Y_T = y_T \mid X_t = i, \theta). \tag{4}$$

Here $\alpha_i(t)$ is the joint probability of observing the returns from time 1 to t and being in state i at time t, given the current parameter estimates, and $\beta_i(t)$ is the probability of observing

the returns from time $t+1$ to T given state i at time t. Both are computed recursively:

$$\alpha_i(1) = \pi_i b_i(y_1),$$
$$\alpha_i(t+1) = b_i(y_{t+1}) \sum_{j=1}^{N} \alpha_j(t)\, a_{ji}, \tag{5}$$
$$\beta_i(T) = 1,$$
$$\beta_i(t) = \sum_{j=1}^{N} \beta_j(t+1)\, a_{ij}\, b_j(y_{t+1}). \tag{6}$$

These recursions can be justified as follows.

1. At $t = 1$, the system is in state i with probability π_i, and $b_i(y_1)$ is the probability of observing the return y_1 given state i. For β we work backwards, so at $t = T$ there are no remaining observations and the probability is 1.

2. At $t+1$, we take the probability of observing y_{t+1} in state i and multiply it by the sum, over all states j, of $\alpha_j(t)$ and the probability of transitioning from j to i. This accounts for every possible path from t to $t+1$. For β, we look at the values at $t+1$ and sum over all states possible at $t+1$, weighted by the probability of moving from the current state to that state and emitting the next observation.

2.3 Parameter Updates (Maximization Step)

To update the parameters we use α and β to construct two auxiliary quantities, γ and ξ, which allow Bayes' theorem to be applied in the updates. The quantity $\gamma_i(t)$ is the probability of being in state i at time t given the full sequence Y and the current parameter estimates, and $\xi_{ij}(t)$ is the probability of being in state i at time t and state j at time $t+1$ given the same information:

$$\gamma_i(t) = \frac{\alpha_i(t)\,\beta_i(t)}{\sum_{j=1}^{N} \alpha_j(t)\,\beta_j(t)}, \tag{7}$$

$$\xi_{ij}(t) = \frac{\alpha_i(t)\, a_{ij}\, \beta_j(t+1)\, b_j(y_{t+1})}{\sum_{k=1}^{N}\sum_{w=1}^{N} \alpha_k(t)\, a_{kw}\, \beta_w(t+1)\, b_w(y_{t+1})}. \tag{8}$$

The parameter updates follow from derivations similar to the EM derivations performed in class. For the transition parameters, with current state i and next state j,

$$\pi_i^* = \gamma_i(1), \qquad a_{ij}^* = \frac{\sum_{t=1}^{T-1} \xi_{ij}(t)}{\sum_{t=1}^{T-1} \gamma_i(t)}. \tag{9}$$

For the emission parameters of state k,

$$\mu_k = \frac{\sum_{t=1}^{T} \gamma_k(t)\, y_t}{\sum_{t=1}^{T} \gamma_k(t)}, \quad \sigma_k^2 = \frac{\sum_{t=1}^{T} \gamma_k(t)\,(y_t - \mu_k)^2}{\sum_{t=1}^{T} \gamma_k(t)}. \tag{10}$$

The expectation and maximization steps are repeated until the parameter estimates converge.

2.4 Viterbi Algorithm

The Viterbi algorithm recovers the most likely sequence of regimes without enumerating every possible state sequence, of which there are 3^{4131}. Instead, it considers the paths that could lead to each state and the probability of arriving there.

The algorithm begins by multiplying the initial distribution π_i by the emission density $b_i(y_1)$, giving the likelihood of observing y_1 and being in state i at $t = 1$. It then proceeds iteratively. For each state at time t, it finds the predecessor state at $t-1$ that maximizes the product of the probability of being in that predecessor state and the probability of transitioning from it to the current state, and multiplies the result by the emission density of y_t. This fills a table whose entries are the probability of reaching each state at time t along the most probable path. During this forward pass the algorithm also records the predecessor from which each state was reached. At time T we select the most probable final state and backtrack through the stored predecessors to recover the full sequence. Formally,

$$P_{t,s} = \begin{cases} \pi_s\, b_s(y_t) & t = 1, \\ \max\limits_{r \in S} \left(P_{t-1,r}\, a_{r,s}\, b_s(y_t) \right) & t > 1, \end{cases} \tag{11}$$

where S is the set of regimes. The recovered sequence is the most probable path of market regimes over the sample period, so the initial goal is achieved using only the price history of the S&P 500.

2.5 Numerical Stability

Because products of many probabilities approach zero, two measures were taken to prevent underflow. In the Baum-Welch procedure, the transition matrix estimates were scaled by the sum of their entries at each time step. In the Viterbi algorithm, the computation was carried out in log space, so that

$$\delta_t(s) = \max_{r \in S} \left[\delta_{t-1}(r) + \log a_{r,s} \right] + \log b_s(y_t), \tag{12}$$

which replaces multiplication with addition and removes the scaling issue.

3 Experimental Results

We fit both a two-state model (bear and bull) and a three-state model (bear, bull, and sideways) in order to examine the effect of the number of regimes. Figures 1 and 2 show the SPY log returns and price path colored by the Viterbi regime sequence, and Table 1 lists the estimated emission parameters.

4 Discussion

The combination of the Baum-Welch and Viterbi algorithms worked well for estimating the market regimes between 2010 and 2026. Comparing the two- and three-state models clarifies the role of the third regime. In the three-state model, the sideways regime typically appears as a boundary between bullish and bearish markets. Removing it produces longer

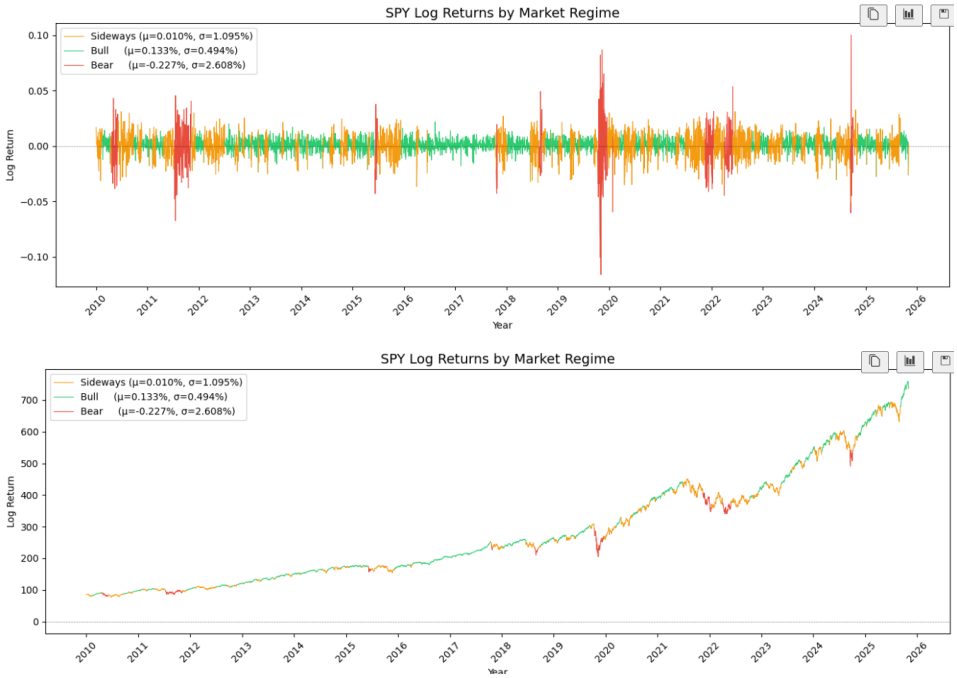


Figure 1: Three-state model. SPY daily log returns (top) and SPY price (bottom), colored by Viterbi regime (bull, bear, sideways).

Model	Regime	μ (%)	σ (%)
Three-state	Bull	0.133	0.494
	Sideways	0.010	1.095
	Bear	−0.227	2.608
Two-state	Bull	0.122	0.601
	Bear	−0.095	1.690

Table 1: Estimated daily emission parameters for each model.

bearish sequences, since the transition period is absorbed into the remaining states and the requirements for each regime are slightly relaxed.

The key features to identify in the figures are the bearish stretch in 2020, when COVID-19 struck, and the bearish market in 2022, when inflation readings were much higher than expected and banks sought to tighten their holdings and reduce market liquidity. Earlier in the series, there was noticeably less movement in prices. As a result, the thresholds for bullish and bearish behavior were less stringent, and the HMM classified some early regions as bearish that would not be considered bearish later in the sample. This is a consequence of fitting a single set of emission parameters to the entire period.

5 Conclusion

Applying the Baum-Welch and Viterbi algorithms to the classification of stock market regimes is a natural fit and yields classifications that appear highly accurate, which is notable given that the only information used was the price history of the market. We first converted prices to log returns. We then estimated the HMM parameters, including the transition matrix, the initial state distribution, and the emission parameters (the mean and variance of a normal distribution for each state), using the Baum-Welch algorithm. Finally, we used the Viterbi algorithm to find the most probable sequence of regimes over the time series.

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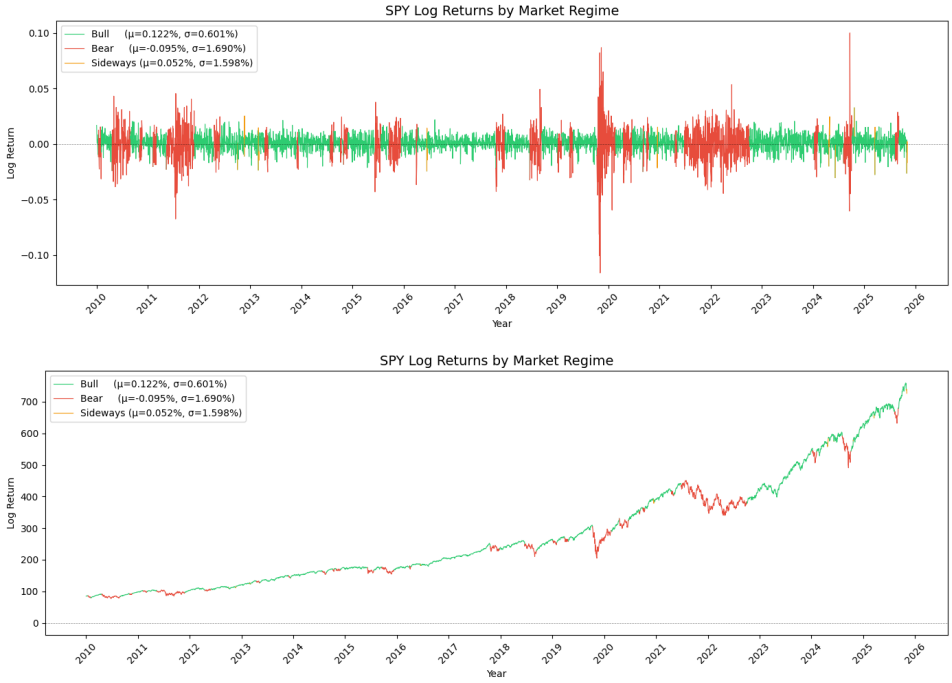


Figure 2: Two-state model. SPY daily log returns (top) and SPY price (bottom), colored by Viterbi regime (bull, bear).